

Joanna Zou

[Google Scholar](#) | [LinkedIn](#) | [Github](#)

Email: jjzou@mit.edu

Research areas Applied statistics, probability, and machine learning, stochastic differential equations, dynamical systems, measure transport, Bayesian inference, Kalman filtering, uncertainty quantification, active learning, risk analysis, signal processing, computational science and engineering.

Research appointments	Massachusetts Institute of Technology	Cambridge, MA
	<i>Postdoctoral Associate</i>	Jun. 2026 - Present
	<i>Graduate Research Assistant</i>	Feb. 2022 - May 2026
	University of Potsdam	Potsdam, Germany
	<i>Visiting PhD Research Fellow in Mathematics</i>	Jun. 2024 - Aug. 2024
	Sandia National Laboratories	Livermore, CA
	<i>Graduate Research Intern</i>	Jun. 2023 - Aug. 2023
	Technical University of Delft	Delft, Netherlands
	<i>Fulbright Research Fellow</i>	Jan. 2021 - Dec. 2021
	NHERI-SimCenter	Berkeley, CA
	<i>Software Development & Research Associate</i>	Aug. 2020 - Dec. 2020
	Stanford University	Stanford, CA

<i>Graduate Research Assistant</i>	Sep. 2018 - May 2020
Arup	San Francisco, CA
<i>Advanced Technology & Research Intern</i>	Jun. 2018 - Aug. 2018

Education	Massachusetts Institute of Technology	Cambridge, MA
	PhD in Computational Science & Engineering	Feb. 2022 – May 2026
	• Advisor: Youssef Marzouk	
	• Thesis: <i>Goal-Oriented Learning of Stochastic Dynamical Systems</i>	
	Stanford University	Stanford, CA
	MS in Structural Engineering	Sep. 2018 – May 2020
	• Advisor: Gregory Deierlein	
	• Stanford Engineering Graduate Fellowship recipient	
	Columbia University	New York, NY
	BS in Civil Engineering	Aug. 2014 – May 2018
	• GPA: 3.95 / 4.00 (<i>magna cum laude</i>)	
	• Top graduate in civil engineering (out of 32)	

Honors and scholarships	SFB 1294 Data Assimilation PhD Research Fellowship	2024
	Fulbright U.S. Student Scholarship	2020
	Marshall Scholarship Finalist	2020

Stanford Engineering Graduate Fellowship	2018
Thornton-Tomasetti Foundation Scholarship	2018
ASCE Arthur Tuttle Fellowship	2018
Columbia George Wendell Memorial Award	2018
Columbia Robert Ridgway Award	2018
ACEC New York Merit Scholarship	2017
Columbia University Henry Michel Scholarship	2015

Publications *Preprints*

[P7] J. Zou, F. Birks, D. Foster, Y. Marzouk. “Stein kernelized molecular dynamics for active learning of interatomic potentials.” *arXiv preprint arXiv:2606.04100*. 2026.

[P6] J. Zou, F. Birks, D. Foster, Y. Marzouk. “Goal-oriented learning of stochastic differential equations using error bounds on path-space observables.” *arXiv preprint arXiv:2063.20467*. 2026.

Journal articles

[J5] J. Zou, E. Lourens, A. Cicirello. “Virtual sensing of subsoil strain response in monopile-based offshore wind turbines via Gaussian process latent force models.” *Mechanical Systems & Signal Processing*. 200, 110488. 2023.

Peer-reviewed conference proceedings

[C4] J. Zou, Y. Marzouk. “Data curation for machine learning interatomic potentials by determinantal point processes.” *ICLR AI for Accelerated Materials Design Workshop*. 2025.

[C3] J. Zou, A. Cicirello, A. Iliopoulos, E. Lourens. “Gaussian process latent force models for virtual sensing in a monopile-based offshore wind turbine.” *Proceedings of the European Workshop on Structural Health Monitoring*, pp. 290-298. 2022.

[C2] J. Zou, D. Welch, A. Zsarnoczay, A. Taflanidis, G. Deierlein. “Surrogate modeling for the seismic response estimation of residential wood frame structures.” *Proceedings of the 17th World Conference in Earthquake Engineering*. 2020.

[C1] Y. Dong, J. Zou, J. Liu, J. Fagert, M. Mirshekari, L. Lowes, M. Iammarino, P. Zhang, H. Noh. “Physics-informed analysis of patient induced structural vibration data for early muscular dystrophy detection.” *Third Workshop on Combining Physical and Data-Driven Knowledge in Ubiquitous Computing (Ubicomp-CPD)*, pg. 525-531. 2020.

**Talks and
presentations**

Invited talks

[T13] “Dimension reduction on Hilbert spaces of drift functions of stochastic differential equations.” *SIAM Conference on Uncertainty Quantification (SIAM UQ)*, Minneapolis, MN. March 2026.

[T12] “Goal-oriented learning of stochastic dynamical systems using error bounds on path-space observables.” *SIAM Conference on Applied Dynamical Systems (SIAM DS)*, Denver, CO. May 2025.

[T11] “Path-space error bounds for the statistical learning of stochastic differential equations.” *University of Potsdam SIAM Chapter Seminar*, Potsdam, Germany. August 2024.

[T10] “Cairn.jl: Enhanced Molecular Dynamics for Active Learning.” *JuliaCon*, Eindhoven, Netherlands. July 2024.

[T9] “Active learning of energy-based models via stochastic Stein variational gradient descent.” *SIAM Conference on Uncertainty Quantification (SIAM UQ)*, Trieste, Italy. February 2024.

[T8] “Adaptive importance sampling for gradient-based dimension reduction in stochastic systems.” *International Conference on Uncertainty Quantification in Computational Science and Engineering (UNCECOMP)*, Athens, Greece. June 2023.

[T7] “Gaussian process latent force models for virtual sensing of offshore wind turbines.” *European Workshop on Structural Health Monitoring (EWSHM)*, Sicily, Italy. June 2022.

Poster presentations

[P6] “Goal-oriented learning and uncertainty quantification of stochastic differential equations.” *MIT Center for the Exascale Simulation of Coupled High Enthalpy Fluid-Structure Interactions (CHEFSI)*, Cambridge, MA. April 2026.

[P5] “Goal-oriented learning of machine learning potentials using error bounds on path-space observables.” *ICMS Workshop on Computational Materials Science and Mathematics at the Particle and Atomistic Scales*, Edinburgh, UK. November 2025.

[P4] “Data curation for machine learning interatomic potentials by determinantal point processes.” *ICLR AI for Accelerated Materials Design Workshop*, Singapore. April 2025.

[P3] “Using error bounds on molecular properties for goal-oriented learning of machine learning potentials.” *MIT Center for the Exascale Simulation of Material Interfaces in Extreme Environments (CESMIX)*, Cambridge, MA. April 2025.

[P2] “Active subspaces for Bayesian inference of machine learning interatomic potential models.” *MIT Center for the Exascale Simulation of Material Interfaces in Extreme Environments (CESMIX)*, Cambridge, MA. Oct. 2023.

[P1] “Bayesian modeling and active learning for molecular dynamics: fitting machine learning interatomic potentials to data.” *EPFL CECAM Workshop on Error Control in First Principles Modeling*, Lausanne, Switzerland. June 2022.

**Academic
service**

Minisymposium organization

“Advances in MCMC Sampling Methods”

SIAM Conference on Uncertainty Quantification (SIAM UQ), 2026.

“UQ for Multiscale Modeling in Computational Chemistry”

SIAM Conference on Uncertainty Quantification (SIAM UQ), 2026.

“Learning with Dynamical Systems and Their Observable Statistics”

SIAM Conference on Applied Dynamical Systems (SIAM DS), 2025.

Peer reviews

International Journal of Uncertainty Quantification (IJUQ) since 2026

Mechanical Systems and Signal Processing (MSSP) since 2024

Data-Centric Engineering (DCE) Journal since 2024

Student organizations

MIT Assoc. of Comp. Engineering and Science Students (ACSES)

Vice President 2023-2025

MIT AeroAstro Graduate Admissions Assistance Program

Mentor 2022-2025

Columbia American Society of Civil Engineers (ASCE)

President 2017-2018

**Teaching
experience**

Stanford Academic Resources Center

Undergraduate and Graduate Course Tutor Summer 2020

Stanford CEE 280: Advanced Structural Analysis

Guest Lecturer Fall 2019

**Professional
memberships**

Society for Industrial and Applied Mathematics (SIAM) 2022 - Present

Earthquake Engineering Research Institute (EERI) 2018-2020

American Society of Civil Engineers (ASCE) 2017 - 2023